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United States Patent	12403720
Kind Code	B2
Date of Patent	September 02, 2025
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Marker pixel wicking apparatus

Abstract

The present disclosure is directed to a marker pixel wicking apparatus that allows a user to create dimensional pixelated art using a method of applying a colored ink marker directly onto the top white surface of a specifically designed substrate. The substrate has been molded specifically with an array of small vertical flat-topped cylinders covering nearly the entire surface.

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Appl. No.:	17/557543
Filed:	December 21, 2021

Prior Publication Data

Document Identifier	Publication Date
US 20220227165 A1	Jul. 21, 2022

Related U.S. Application Data

us-provisional-application US 63185627 20210507
us-provisional-application US 63128427 20201221

Publication Classification

Int. Cl.: G09B11/04 (20060101); B43K8/06 (20060101); B44D2/00 (20060101)

U.S. Cl.:

Field of Classification Search

USPC: None

References Cited

U.S. PATENT DOCUMENTS

Patent No.	Issued Date	Patentee Name	U.S. Cl.	CPC
4212393	12/1979	Lenkoff	434/84	A63H 33/22
2018/0065265	12/2017	Volzer	N/A	B26D 1/04

FOREIGN PATENT DOCUMENTS

Patent No.	Application Date	Country	CPC
203580428	12/2013	CN	N/A

OTHER PUBLICATIONS

Product Information, ScrewFix, 2022, “Fix-O-Moll Natural Round Self-Adhesive Parquet Gliders”, screwfix.com [online], Available from: (See Additional information) [Jan. 6, 2022] (2 pgs) See relevant figures [https://www.screwfix.com/p/fix-o-moll-natural-round-self-adhesive-parquet-gliders-22-x-22mmm-16-pack/544KF?](https://www.screwfix.com/p/fix-o-moll-natural-round-self-adhesive-parquet-gliders-22-x-22mmm-16-pack/544KF?tc=GT8&ds_kid=92700048793290424&ds_4I=1249413&gclid=EAlalQobChMI9auHye6L-AIVC91RChliuQunEAkYBiABEGltBvD_BwE&gclsrc=aw.ds)

[tc=GT8&ds_kid=92700048793290424&ds_4I=1249413&gclid=EAlalQobChMI9auHye6L-AIVC91RChliuQunEAkYBiABEGltBvD_BwE&gclsrc=aw.ds](https://www.screwfix.com/p/fix-o-moll-natural-round-self-adhesive-parquet-gliders-22-x-22mmm-16-pack/544KF?tc=GT8&ds_kid=92700048793290424&ds_4I=1249413&gclid=EAlalQobChMI9auHye6L-AIVC91RChliuQunEAkYBiABEGltBvD_BwE&gclsrc=aw.ds). cited by applicant

Amazon Product Information, 2022, “Water Drawing Mat-Kids Aqua Water Doodle Mat Toy”, Amazon.com, [online], Available from: [https://www.amazon.co.uk/Water-Drawing-Mat-Kids-Doodle/dp/B07PCHQ1SY/ref=sr_1_1?](https://www.amazon.co.uk/Water-Drawing-Mat-Kids-Doodle/dp/B07PCHQ1SY/ref=sr_1_1?crid=2B5CCVBCS3CAH&keywords=water+drawing+mat+-+kids+aqua+water+doodle+mat+toy&qid=1655997712&srefix=water+drawing+mat+-+kids+aqua+water+doodle+mat+toy%2Caps%2C147&sr=8-1)

[crid=2B5CCVBCS3CAH&keywords=water+drawing+mat+-+kids+aqua+water+doodle+mat+toy&qid=1655997712&srefix=water+drawing+mat+-+kids+aqua+water+doodle+mat+toy%2Caps%2C147&sr=8-1](https://www.amazon.co.uk/Water-Drawing-Mat-Kids-Doodle/dp/B07PCHQ1SY/ref=sr_1_1?crid=2B5CCVBCS3CAH&keywords=water+drawing+mat+-+kids+aqua+water+doodle+mat+toy&qid=1655997712&srefix=water+drawing+mat+-+kids+aqua+water+doodle+mat+toy%2Caps%2C147&sr=8-1) [Accessed Jan. 6, 2022] (8 pgs). cited by applicant

Combined Search and Examination Report dated Jun. 9, 2022 received in related UK Patent Application GB2118687.9 filed on Dec. 21, 2021 (8 pgs). cited by applicant

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Background/Summary

CROSS-REFERENCE TO RELATED APPLICATIONS (1) This application claims the benefit of U.S. Provisional Application No. 63/185,627, filed May 7, 2021 and U.S. Provisional Application No. 63/128,427, filed Dec. 21, 2020, the contents of each of which are incorporated herein by reference.

FIELD

(1) The present disclosure is directed to a marker pixel wicking apparatus that allows users to create dimensional pixelated art using a method of applying colored ink from a marker directly

onto a blank white surface featuring small vertical cylinders.

BACKGROUND

(2) Blank coloring substrates provide a marking surface for users to create limitless marker artwork, but do not provide boundaries for guiding marker ink application for segmented or pixelated art effects.

SUMMARY

(3) The present disclosure is directed to a marker pixel wicking apparatus that allows a user to create pixelated art using a simple method of applying safe marker ink onto a specially designed 3D surface. The user will not need to use heat while being supervised by an adult as most competitive products require.

(4) Also provided herein is system for creating pixelated marker art using a porous substrate and frame structure and any number of fluid ink marking instruments.

(5) The present disclosure is also directed to method for creating pixelated artwork on a prepared substrate where a user contacts a fibrous nib of a marking device against a fibrous protrusion on a pixelated substrate. The pixelated substrate may also be secured within a pixel frame structure that further isolates individual porous segments of the underlying substrate.

(6) More specifically, the present disclosure is directed to a marker wicking apparatus comprising a planar base made from a first material, at least one wicking protrusion extending outward from the planar base, the at least one wicking protrusion being made from a second material, the at least one wicking protrusion comprising a wicking surface; wherein the first material and the second material may be the same or different; and wherein a wicking ability of the first material is less than a wicking ability of the second material.

(7) A further aspect of the present disclosure is a marker wicking apparatus comprising a hydrophobic base; and a plurality of hydrophilic columnar wicking posts coupled to and extending upward from the hydrophobic base, the wicking posts configured to receive and wick a donor solution throughout the post, wherein the hydrophobic base is configured to resist wicking the donor solution wicked through the wicking post.

(8) Other objects and features will be in part apparent and in part pointed out hereinafter.

Description

BRIEF DESCRIPTION OF THE DRAWINGS

(1) FIG. 1 is a cross-sectional view of one pixel of an embodiment of the marker pixel wicking apparatus.

(2) FIG. 2A is a perspective view of the marker pixel wicking apparatus with cylindrical columns.

(3) FIG. 2B is an exploded perspective view of the marker pixel wicking apparatus of FIG. 2A.

(4) FIG. 3A is a perspective view of a marker pixel wicking apparatus without an outside frame.

(5) FIG. 3B is an exploded perspective view of the marker pixel wicking apparatus of FIG. 3A.

(6) FIG. 4 is a top view of an embodiment of the marker pixel wicking apparatus.

(7) FIG. 5A is a side view of an embodiment of the wicking apparatus.

(8) FIG. 5B is a side view of another embodiment of the wicking apparatus.

(9) FIG. 6A is a top perspective view of a wicking apparatus with a frame structure and artwork.

(10) FIG. 6B is an exploded perspective view of the wicking apparatus and frame structure of FIG. 6A.

(11) FIG. 7 is a top plan view of a porous marker wicking apparatus having rectangular vertical wicking posts, in accordance with one embodiment of the present disclosure.

(12) FIG. 8A depicts a top plan view of the porous marker wicking apparatus of FIG. 7.

(13) FIG. 8B depicts a side plan view of the porous marker wicking apparatus of FIG. 7.

(14) FIG. 9 is a perspective view of a vertical wicking post with sharp edges.

- (15) FIG. 10. is a perspective view of a vertical wicking post with rounded edges.
- (16) FIG. 11 is a perspective view of a porous marker wicking apparatus having removable rectangular vertical wicking posts in accordance with one embodiment of the present disclosure and illustrating use of tweezers to insert a vertical wicking post; and a close-up view of a marker coloring a vertical wicking post.
- (17) FIG. 12A is a side view of an eraser tool in accordance with one embodiment of the present disclosure.
- (18) FIG. 12B is a perspective view of the porous marker wicking apparatus of FIG. 7 with a sponge eraser tool in accordance with one embodiment of the present disclosure.
- (19) FIG. 12C is a close-up of the eraser tool erasing color from a vertical wicking post.
- (20) FIG. 13 is a perspective view of the porous marker wicking apparatus of FIG. 7 with an art guide with handles in accordance with one embodiment of the present disclosure and illustrating use of a marker to color a vertical wicking post.
- (21) FIG. 14 is a perspective view of the porous marker wicking apparatus of FIG. 7 with an art guide without handles in accordance with one embodiment of the present disclosure and illustrating use of a marker to color a vertical wicking post; a close-up view of the marker coloring a vertical wicking post; and a top plan view showing the pixel art that is created on the porous marker wicking apparatus.
- (22) FIG. 15A is a top plan view of the porous marker wicking apparatus of FIG. 7 with an art guide without handles and having a central opening for the wicking apparatus in portrait orientation.
- (23) FIG. 15B is a top plan view of the porous marker wicking apparatus of FIG. 7 with an art guide without handles and having a central opening for the wicking apparatus in landscape orientation.
- (24) FIG. 16A is a perspective view of the porous marker wicking apparatus of FIG. 7 with a light board.
- (25) FIG. 16B is an exploded perspective view of the light board and porous marker wicking apparatus of FIG. 16A
- (26) FIG. 16C is a side view of the porous marker wicking apparatus and light board of FIG. 16A.
- (27) FIG. 17 is a top plan view of a modular frame for use with a porous marker wicking apparatus in accordance with one embodiment of the present disclosure.
- (28) FIG. 18 is an exemplary kit containing porous marker wicking apparatuses of FIG. 7, adhesive, image guide, markers, an art guide of FIG. 15A and matte framing.
- (29) Corresponding reference characters indicate corresponding parts throughout the drawings.

DETAILED DESCRIPTION

(30) The present disclosure is related to a marker pixel wicking apparatus that allows a user to create dimensional pixelated art using a method of applying a colored ink marker directly onto the top white surface of a specifically designed substrate. The substrate has been molded specifically with an array of small vertical flat-topped cylinders covering nearly the entire surface. Any color marker nib is placed directly onto the cylinders' top surface to activate the ink transfer to create a colorful image. A user can select any desired colors and/or placement across the array of cylinders to create a colorful image. An unlimited amount of images can be created simply by arranging the colors across the array of white cylinders. The Pixel Art surface may be washable allowing repeated use if desired. Washing the surface using clean water removes any transferred color returning the surface to its original blank white color. The wicking substrate is typically molded or co-molded using two separate materials that will allow the protruding vertical cylinders to easily absorb the marker ink using a porous material. Selective laser sintering can also be used. This allows for brighter, more vibrant colors in the finished pixel art. The base layer below the cylinders is molded in a denser, less absorbing material to limit the spread of color, mitigating any color contamination.

(31) In the embodiments of FIGS. 1-6, various views of the marker pixel wicking apparatus are shown. In some aspects, the substrate surface is a wicking, porous material similar to that of a marker nib. In further aspects, the pixel wicking substrate has a first porosity proximate the outer surface of the substrate, and a second porosity proximate the base plate, such that the flow of wicked ink into the contacted wicking substrate surfaces does not flow below the higher density, second porosity structure. As further shown in the examples of FIGS. 1-6, the staggered, honeycomb orientation of the individual “pixel” columnar segments of the porous substrate indicate individual components for coloring with a washable ink, such as a washable, water-based ink from a broadline marker. The base and/or frame structure may be configured to support the wicking porous substrate along one or more surfaces, such as the bottom surface along a solid panel parallel to the substrate, and a top surface perforated with openings corresponding to the individual pixel protrusions of the top surface of the wicking substrate.

(32) Turning now to FIGS. 7-11, another embodiment of the marker pixel wicking apparatus is generally indicated at reference numeral 110. Wicking apparatus comprises a base 112 and a plurality of vertical wicking posts (protrusions) 114 extending upwardly along the upper surface of base 112. Vertical wicking posts 114 can be any desired shape, such as cylindrical (see FIGS. 2-6), rectangular (see FIGS. 7-11), or any other suitable shape known to those skilled in the art. That is, the cross-sectional shape of the vertical wicking post 114 can be a circle, square, rectangle, oval, star, heart, polygon, or any desired shape or combination thereof. In certain embodiments, wicking posts 114 can have a width w1 and/or depth d1 of less than about 0.2 inches (e.g., 0.15 inches), for example, from about 0.1 inches to about 0.2 inches, in some embodiments about 0.15 inches. In the embodiment shown in FIG. 8, the vertical wicking posts 114 have a square cross-section such that the width w1 and/or depth d1 are equal. In the exemplary embodiment shown in FIG. 10, the edges of wicking posts 114 are rounded. Alternatively, in the exemplary embodiment shown in FIG. 9, the edges of wicking posts 114 are sharp (i.e., not rounded).

(33) In certain embodiments, wicking posts 114 can have a height h1 (measured from the top surface of the base 112 to the top surface of the wicking post 114) of from about 0.1 inches to about 0.2 inches, in some embodiments about 0.12 or about 0.15 inches. In various aspects, wicking posts 114 are uniformly spaced across base 112 to form evenly spaced columns and rows of wicking posts 114. Wicking posts 114 are also uniformly spaced from each other. In some embodiments the distance between wicking posts 114 can be less than about 0.1 inches, for example about 0.06 inches, or from about 0.03 inches to about 0.1 inches. The wicking posts 114 can generally comprise perimeter wicking posts 114a around the perimeter of the base 112 and central posts 114b in the center area of the base 112. Central posts 114b are generally separated from adjacent posts 114, whether a perimeter or another central post, by a first distance (e.g., the distances recited above). The distance between the perimeter posts 114a and the nearest edge(s) of the base 112 is about one half of the first distance. This allows for a combination of multiple apparatuses 110 without any visible gap between distinct panels.

(34) A sufficient number of wicking posts 114 are included on base 112 to generally cover its surface area with the wicking posts 114 positioned substantially equally apart across the surface, the number of wicking posts 114 varying based on the size and dimensions of base 112. For example, in the embodiment shown in FIG. 8, the surface area of the base is approximately 20 inches (e.g., approximately 5 (4.83) inches by 4 (3.99) inches). In this embodiment, the base 112 includes 19 wicking posts 114 in a column orientation and 23 wicking posts 114 in a row orientation.

(35) In another embodiment, and as shown in FIG. 11, wicking posts 114 can be removable from base 112. In this embodiment, base 112 includes a plurality of wells 116 sized and shaped to receive wicking posts 114. Wells 116 are deep enough to allow insertion and stable retention of the wicking posts 114 but do not fully encapsulate wicking posts 114. For example, wells 116 may be from about 0.05 inches to about 0.1 inches deep, such as 0.08 inches deep, compared to an overall height of individual wicking posts 114 measured from the base of the well 116 to the top surface of

wicking post **114** of from about 0.1 inches to about 0.3 inches (e.g., approximately 0.2 inches). (36) Wicking posts **114** are made of a substrate that is molded or co-molded using materials that allow the wicking posts **114** to absorb a donor solution (e.g., marker ink). The apparatus **110** itself can be formed by selective laser sintering, molding, or any other appropriate means known in the art. Thus, wicking posts **114** may be made from a porous and/or hydrophilic material to facilitate wicking or absorption of the donor solution. In contrast, base **112** is typically made of (molded in) a denser/less porous and/or hydrophobic material such as a molded plastic to inhibit the spread of the donor solution through base **112**. Alternatively, both the base **112** and wicking posts **114** can be made from the same material. Typically, when this is the case, the base **112** is treated with a hydrophobic substance and/or the wicking posts **114** are treated with a hydrophilic substance. In various aspects, the porosity of the wicking posts **114** is higher than the porosity of the base **112**. In some aspects, the porosity of the base **112** is increased in order to improve the washability of the apparatus **110**. The increased porosity can be across the entire base **112** or only sections thereof (for example, the upper surface of base **112**) in order to reduce flow of donor solution deeper into base **112**, which in turn makes it easier to clean apparatus **110** and reduces the chance of donor solution leaching into the base **112** or reappearing in the base **112** after being washed. For example, the wicking posts **114** can have a porosity of from about 20 microns to about 40 microns. The porosity of the base **112** is less than or equal to the porosity of the wicking posts **114** and can be, for example, from about 0 microns to about 40 microns.

(37) As shown in FIG. **11**, the apparatus **110** may also be provided with a pair of tweezers **118** to allow a user to easily grasp the individual wicking posts **114** for placing them into and removing them from well **116**. It may also optionally and independently be provided with one or more markers **120** to be used in coloring the wicking posts **114**.

(38) In some embodiments, one or more surfactants are incorporated in the top (surface) portion of wicking posts **114** in order to enhance absorption of the donor solution into the wicking posts. In these embodiments, repeated use and cleaning of the apparatus may cause the surfactants to be washed out of the wicking post **114**. Additional surfactant to replenish the lost surfactant can be provided with the apparatus for adding to the top portion of wicking post **114**. In various embodiments, additional surfactant is adding to the wicking posts **114** during standard washing with soap (e.g., dish soap), acting as the surfactant, and water. After use of the apparatus **110**, the user can simply wash the apparatus **110** with soap and water, thereby washing away any color on the posts **114** and delivering additional surfactant to aid in the wicking action.

(39) A delivery device or system could additionally be provided for evenly applying the additional surfactant such as via a brush, a dropper, or a surfactant marker. In various aspects, the additional surfactant is provided in a surfactant marker (e.g., a marker having a reservoir and a saturated nib comprising the surfactant). The surfactant can be replenished in the top of the wicking post **114** by contacting the marker nib with the wicking post for a sufficient amount of time to saturate the top surface of the wicking post with surfactant. In other embodiments, the top portion of wicking posts **114** may not originally contain a surfactant solution, but optionally, the user can apply surfactant solution if desired to wicking posts **114** in the manner described above.

(40) In various embodiments, more than one pixel wicking apparatus **110** can be combined to create a larger drawing surface. Any desired number of pixel wicking apparatuses **110** can be placed side-by-side to enlarge the drawing surface. The pixel wicking apparatuses **110** can be joined by any appropriate method known in the art. For example, multiple pixel wicking apparatuses **110** can be joined by adhering an adhesive sheet to the underside (i.e., bottom) of the bases of the apparatuses **110**. Wicking apparatuses **110** can also be joined using fusible plastic beads placed on the bottom face of the apparatuses **110** (either across the majority of the surface area of the bottom face or on joining edges) and fused together using heat, water, or any other appropriate means. Wicking apparatuses **110** can also be placed side-by-side on a substrate and secured by any known means to said substrate using, for example, adhesive (glue dots, glue, hot

glue, tape, and loop-type fasteners etc.).

(41) The wicking apparatus **110** can also be treated with a protectant that provides UV protection, fading or ink transfer protection (e.g., a fixative), or any other desired characteristic. The protectant can be in a spray, liquid, or solid form. A fixative seals in the image created on the wicking apparatus by closing the open pores of the wicking posts. The fixative can be a resin, spray, or any other suitable substance and application method known in the art.

(42) Thus, provided herein is a method of drawing using the pixel wicking apparatus. A user places a donor solution source (e.g., a saturated marker nib) onto the top surface of at least one vertical wicking posts for a sufficient amount of time to saturate the top surface of the vertical wicking post with the donor solution. This step can be repeated any desired number of times on different vertical wicking posts and with different donor solution sources (e.g., saturated marker nibs of differing ink colors) to produce any desired image. In various embodiments, when the drawing is completed, the user can seal the drawing using a fixative in order to create a waterproof barrier and/or reduce the fading of ink. Without the fixative in place, the donor solution can be washed from the vertical wicking posts to create a clean, uncolored drawing surface. Washing with clean water removes any transferred donor solution to return the vertical wicking posts to their original color (e.g., white) for re-use in creating a new pixel artwork.

(43) Turning now to FIG. **12**, also provided herein is a sponge eraser tool **122** for use with the pixel wicking apparatus **110**. As illustrated, the eraser **122** comprises an absorbent sponge **128** having the same geometry (e.g., same cross-sectional dimensions) as the individual vertical wicking posts **114**. For example, in embodiments where the vertical wicking posts **114** are cylindrical, the sponge **128** will have a circular cross-section that is the same diameter as the vertical wicking posts **114**. In other embodiments where the vertical wicking posts **114** are square or rectangular, the sponge **128** will have a square or rectangular cross-section that has the same width and depth as the vertical wicking post **114**. In general, the absorbent sponge **128** is made from a material that has a greater absorbency than that of the wicking post's **114** porous material. For example, in some aspects, the sponge **128** may be made from polyvinyl alcohol (PVOH) or cellulose. The absorbent sponge can be partially inserted into a barrel **126** for ease of use, such that one end of the sponge **128** is secured within the barrel **126** and an opposite end of the sponge **128** is exposed for use.

(44) In some embodiments, pixel wicking apparatus **110** can be provided with and/or used in conjunction with pixel templates that serve as guides to assist the user in creating specific artwork images. For example, the pixel templates may be provided in the form of printed sheets of paper where a top view of the pixel wicking apparatus is printed on each sheet displaying the tops of the vertical wicking posts. Each template may have a different image (e.g. design or picture) printed on it wherein the image is created by the specific arrangement of colored vertical posts on the sheet. This template may identify different colors to be used in creating the image, and may identify the specific position/location of each vertical wicking post that is colored to create the design or picture with reference to a reference number or other position locator. In other embodiments, the pixel templates may be provided in electronic form such as by accessing the templates on-line via a website. In this embodiment, a library of different images can be made available for user selection, wherein the user can select the desired image and then print or display the template for that image on a screen for use as a guide in making a similar image with the pixel wicking apparatus. In other embodiments, custom templates may be made by or for the user that provide a guide for making a specific image that the user desires to create. For example, a custom template may be made by using a software application that functions to receive an image (such as by uploading a scanned drawing or photograph) and converts or translates that image into a pixelated map. In this way, a user can recreate a desired image with the map produced by the software. The pixelated map can also indicate appropriate marker colors to recreate the image. The software may also generate a guide to provide the user the ability to translate the image to specific marker colors. The pixelated map and/or the guide may be printed by the user or referenced on-line for use in creating the image

with the pixel wicking apparatus. Thus, a user uploads an image desired to be recreated in pixel art to the software. The user then crops and fits the image to the size of base of the pixel wicking apparatus. Once this step is complete, the software creates a guide that can be optionally printed by the user or viewed on screen. The guide provides codes and colors to match available markers and indicates which wicking post receives which color to recreate the uploaded image.

(45) Turning now to FIGS. **13** and **14**, another aspect of the present disclosure is an art guide generally indicated at reference numeral **130**. The art guide **130** comprises a base **132** having an opening **136** sized and shaped to receive the base **112** of the pixel wicking apparatus **110**. In the embodiment illustrated in FIGS. **13** and **14**, the opening **136** comprises a slot on one outside edge **134** of the base **132** (e.g., the lower edge). In other embodiments, such as those illustrated in FIGS. **15A** and **15B**, the opening **136** can be located in a central portion of the base **132** such that the perimeter of the opening **136** abuts the outer perimeter of the base **112** of the pixel wicking apparatus **110**. In embodiments where the height and width dimensions of the apparatus **110** differ, the art guide **130** can be in a portrait orientation (FIG. **15A**), a landscape orientation (FIG. **15B**), or interchangeable. In various embodiments, the art guide **130** can also include at least one handle **138**. In the illustrated embodiment, the base **132** defines two openings **140** on opposing sides that comprise two handles **138**. Another handle **138** is located near the top edge of base **132** in the illustrated embodiment. The top edge handle **138** protrudes from base **132**, allowing a user to easily pick up the art guide **130** from a flat surface. The base **132** also includes pixel indicia **142**, such that each row and/or column of vertical wicking posts is individually indicated. For example, in the illustrated embodiment, each column of vertical wicking posts is marked by a distinct letter **142a** and each row of vertical wicking posts is marked by a distinct number **142b**, although it will be understood that any number of identification systems can be used. In some aspects, the art guide **130** is used in conjunction with the above-described pixel template (printed, on-line or custom pixelated map and guide), where the pixel template provides a color for an indicated vertical wicking post **114** using the art guide indicia (e.g., **A1** in the illustrated embodiment), providing a user reference points to recreate a desired image.

(46) Turning now to FIGS. **16A** to **16C**, also provided herein is a light-board pixel system generally indicated at reference numeral **230**. The light-board generally comprises a drawing surface **232** and a light source (not shown) operable to emit light. The drawing surface **232** can comprise a transparent or translucent sheet-like large surface area optical waveguide to mix the wavelengths of light emitted from light emitting diodes (LEDs) to uniformly illuminate the drawing surface with the light. The LEDs can be edge-mounted. The light-board also contains a power switch that supplies power to the LEDs when in the “on” position. In operation, a user places an image (or image guide) **236** she wishes to recreate on top of the light board, with the light board in the “on” position. Then, a wicking apparatus **110** can be placed on top. It is generally preferable for the wicking apparatus base to have a transparent or semi-transparent base **112** when used in this application. In this way, the user can see the color of the picture surrounding the pixel illuminated through the apparatus base by way of the light board and determine appropriate colors to use in order to recreate the picture.

(47) Turning now to FIG. **17**, also provided herein is a modular frame generally indicated at reference numeral **210**. Modular frame **210** comprises a plurality of modular panels **212**. Modular panels **212** comprise a base portion **214** having at least one joining edge **218** and optionally, a framing portion **220**. The framing portion **220** is raised and is located at the perimeter of the modular panel **212**. The presence or absence of a raised edge (framing portion **220**) is determined by the panel's **212** anticipated location in the assembled modular frame **210**. For example, a panel **212** anticipated to be located at the corner of the frame **210** will have two raised edges at the corner of the assembled frame **210**, a panel **212** located at a non-corner perimeter will have one raised edge at the edge of the assembled frame **210**, and a panel **212** located in a central non-perimeter location will have no raised edges. Modular panels **212** can be joined by any means known in the

art. In the illustrated example, modular panels are joined via a mating connection using a mating piece **224**. In this way, each modular panel has at least one cut-out **226** on each joining edge **218**. A mating piece **224** is inserted at the aligned cut-outs **226** of the two modular panels **212** to secure the modular panels **212** together. In this way, mating piece **224** is sized and shaped to be received within the aligned cut-outs **226** of two modular panels **212**. Once the full modular frame **210** is assembled, a pixel wicking apparatus (or multiple pixel wicking apparatuses, as desired) can be placed within the frame **210**. A mat can also be placed around the wicking apparatuses. In other embodiments, a standard frame can also be used.

(48) Various kits including components described herein are also envisioned by the present disclosure. For example, a first kit includes at least one wicking apparatus as described herein (e.g., one, two, three, four, etc.) and at least one marker for use with the apparatus (e.g., at least two, at least five, etc.). The kit may also include a sponge eraser tool, adhesive, modular framing components, art guides, and inspiration or guide pictures to assist a user in creating a design. The kit illustrated in FIG. **18** contains a plurality of wicking apparatuses **110** (e.g., two, four, or six wicking apparatuses), a plurality of writing instruments **120**, more specifically markers, an art guide **130**, an adhesive component **180**, an image guide **182**, and matte framing **184**. In the illustrated embodiment, the matte framing comprises single sheet of substrate having perforations or other guides thereon that allow a user to punch out the desired size of framing. The kit may include some or all of these components, or additional components as necessary.

(49) Having described the invention in detail, it will be apparent that modifications and variations are possible without departing from the scope of the invention defined in the appended claims.

(50) When introducing elements of the present invention or the preferred embodiments(s) thereof, the articles “a”, “an”, “the” and “said” are intended to mean that there are one or more of the elements. The terms “comprising”, “including” and “having” are intended to be inclusive and mean that there may be additional elements other than the listed elements.

(51) In view of the above, it will be seen that the several objects of the invention are achieved, and other advantageous results attained.

(52) As various changes could be made in the above products and methods without departing from the scope of the invention, it is intended that all matter contained in the above description and shown in the accompanying drawings shall be interpreted as illustrative and not in a limiting sense.

Claims

1. A kit for creating pixel artwork or designs, the kit comprising: a marker wicking apparatus comprising: a planar base; a plurality of wicking protrusions extending outward from the planar base, the plurality of wicking protrusions comprising a wicking surface; and wherein a wicking ability of the planar base is less than a wicking ability of the plurality of wicking protrusions; and at least one template, the at least one template displaying an image that can be recreated on the marker wicking apparatus by coloring a specific arrangement of one or more of the plurality of wicking protrusions, the template identifying a specific position and a specific color for each of the one or more of the plurality of wicking protrusions to be colored to recreate the image.
2. The kit of claim 1, wherein each of the plurality of wicking protrusions has a cross-sectional shape of a circle, a square, a rectangle, an oval, a star, a heart, a polygon, or a combination thereof.
3. The kit of claim 1, wherein each of the plurality of wicking protrusions is oriented generally perpendicular to the planar base.
4. The kit of claim 1, wherein the porosity of the planar base is less than the porosity of the plurality of wicking protrusions.
5. The kit of claim 1, wherein each of the plurality of wicking protrusions is coated or saturated with a surfactant on the wicking surface.
6. The kit of claim 1, wherein a top surface of each of the plurality of wicking protrusions

comprises the at least one wicking surface, and the top surface is generally planar and parallel to the planar base.

7. The kit of claim 1, wherein the planar base and the plurality of wicking protrusions are integrally molded.

8. The kit of claim 1, wherein the plurality of wicking protrusions are releasably coupled to the planar base.

9. The kit of claim 8, wherein the planar base defines at least one well sized and shaped to releasably receive the plurality of wicking protrusions.

10. The kit of claim 1, wherein the plurality of wicking protrusions comprises at least about 10, at least about 20, at least about 30, at least about 50, at least about 100, or at least about 200 wicking protrusions.

11. The kit of claim 10, wherein the wicking protrusions are evenly spaced and distributed on the planar base.

12. The kit of claim 11, wherein the plurality of wicking protrusions comprises perimeter protrusions located around the perimeter of the apparatus and central protrusions, wherein the central protrusions are spaced a first distance from an adjacent perimeter protrusion or adjacent central protrusion, and wherein the perimeter protrusions are spaced half of the first distance from an edge of the planar base.

13. The kit of claim 1, wherein each of the plurality of wicking protrusions has a porosity of from about 20 microns to about 40 microns.

14. The kit of claim 13, wherein the planar base has a porosity equal to or less than the porosity of each of the plurality of wicking protrusions.

15. The kit of claim 1, further comprising at least one writing instrument containing a donor solution configured to wick in at least one of the plurality of wicking protrusions.

16. The kit of claim 15, wherein the at least one writing instrument is at least one marker.

17. The kit of claim 1 further comprising an art guide, the art guide comprising a planar base, the planar base defining an opening thereon sized and shaped to receive the planar base of the marker wicking apparatus, wherein the planar base includes indicia configured to identify one wicking protrusion or a group of wicking protrusions of the plurality of wicking protrusions.

18. The kit of claim 17, wherein the indicia comprise column indicia and row indicia, wherein the column indicia are spaced near a horizontal edge of the opening and are configured to indicate to a user which column a wicking protrusion is in, and wherein the vertical indicia are spaced near a vertical edge of the opening and are configured to indicate to a user which row a wicking protrusion is in.

19. The kit of claim 1, wherein the base is formed from a hydrophobic material and the plurality of wicking protrusions are formed from a hydrophilic material.

20. The kit of claim 1, wherein the base and the plurality of wicking protrusions are formed from the same material.

21. The kit of claim 20, wherein the plurality of wicking protrusions have a higher porosity than the base.

22. The kit of claim 20, wherein the base is treated with a hydrophobic substance.

23. The kit of claim 20, wherein the plurality of wicking protrusions are treated with a hydrophilic substance.
